Summary

The dataset contains hourly outage records across 83 Michigan counties from April to June 2023, along with weather features. The outage distribution is highly skewed, with over 70% zeros and extreme spikes. A log transformation was applied to stabilize variance. We evaluate ARIMA, GRU, LSTM, and Transformer models. Performance is measured using MSE, MAE, and R^2. GRU and LSTM outperform ARIMA, while Transformer underperforms overall. GRU achieves the best overall ranking. Using model predictions, we recommend a greedy-based placement of 5 power stations focusing on high-risk counties. Infrastructure improvements are also necessary for high-demand counties.

Exploratory Data Analysis

The dataset contains hourly records for 83 Michigan counties from April 1 to June 30, 2023 along with 109 weather features. The target variable (hourly outage count) is highly skewed: about 70.5% of values are zero, the median is 0 while the mean is 45.25, skewness is 26.37, kurtosis is 947.23, and the maximum reaches 23,346. Because raw values are difficult to model, we apply a log(1+ p) transform, which yields a distribution that remains zero-heavy but is easier to learn (Figure 1).

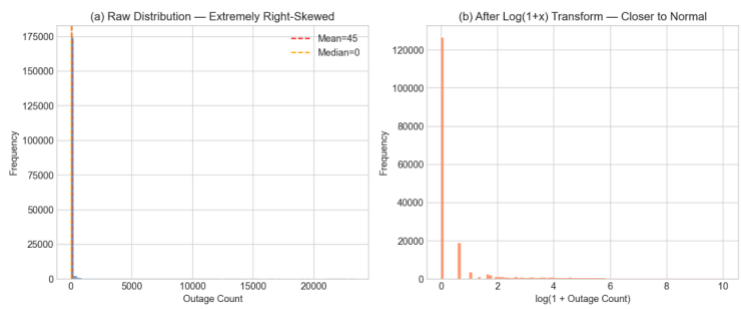

Figure 1 Hourly outage distribution: raw values vs. after log(1+x) transform

Temporally, outages are dominated by a few weather events rather than stable daily cycles. The 24-hour moving average shows minor clusters in April and May and one large spike on June 26, 2023, reaching about 70,000 outages (Figure 2). The hour-of-day mean is nearly flat (peak/low ratio 1.40x), with maxima slightly elevated between 06:00-16:00. The apparent day-of-week peak on Mondays is an artifact of the June 26 event, not a true weekly pattern.

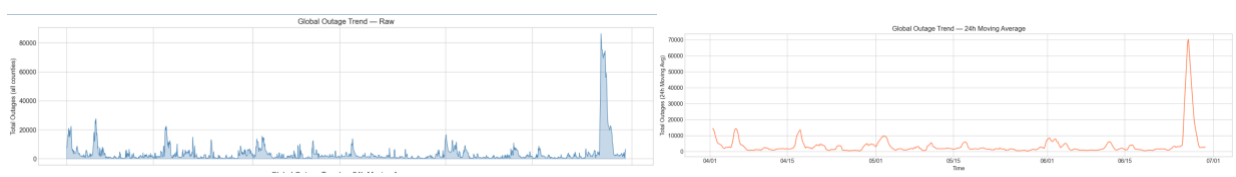

Figure 2 Total outages over time (raw and 24-hour moving average)

Spatially, county-level correlations are weak. Among the top 20 counties, the mean pairwise Pearson correlation is 0.106, with only 7 of 190 pairs above 0.5 (Figure 3). Aside from a small cluster (26125/26163/26099/26161), most counties are nearly independent, motivating separate models per county.

Weather–outage relationships are also weak. The top Spearman correlations (e.g., lai, cape_1, blh, orog) are only around $|\rho| \approx 0.10$–0.13 (Figure 4). Splitting outages into zero vs. nonzero groups yields similarly small effect sizes (mean $|d| = 0.088$, max 0.327), suggesting no need for a separate extreme-event model. However, strong multicollinearity exists among

weather variables: 49 feature pairs have |r| > 0.9 across 38 features (Figure 5), including near-duplicate variables such as pressure (r = 0.9997) and radiation components (r = 0.9972).

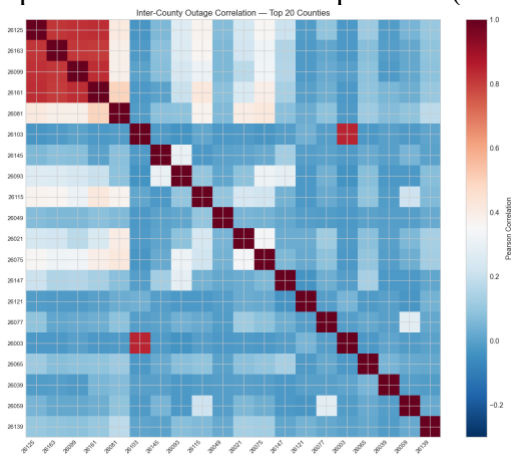

Figure 3 Outage correlation between top 20 counties

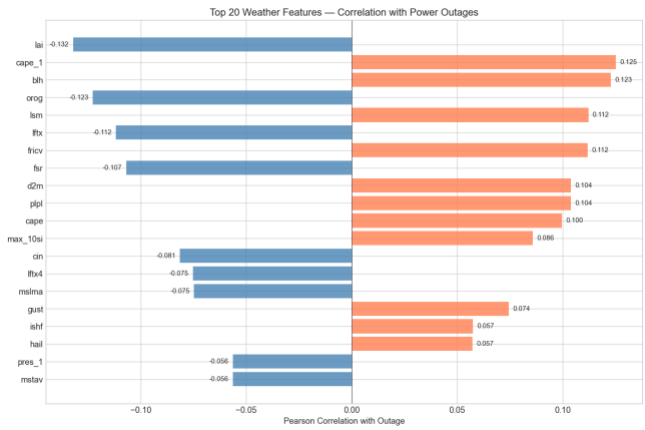

Figure 4 Top 20 weather features vs. outage (Pearson)

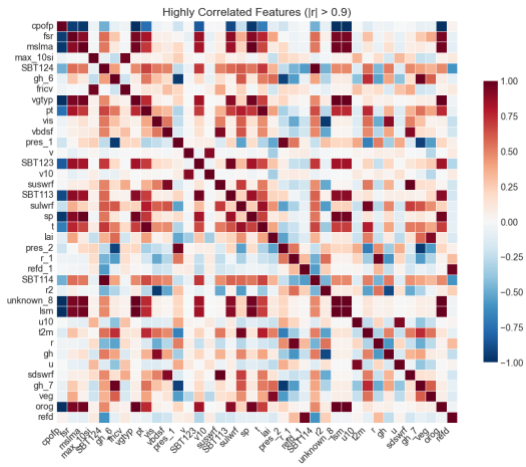

Figure 5 Correlation heatmap of the 38 features

Models

Introduction

We evaluate four models for power outage prediction: Autoregressive Integrated Moving Average (ARIMA) (Box & Jenkins, 1970), Gated Recurrent Unit (GRU) (Cho et al., 2014), Long Short-Term Memory (LSTM) (Hochreiter & Schmidhuber, 1997), and Transformer (Vaswani et al., 2017). These models represent both classical statistical methods and modern deep learning approaches for time-series forecasting.

ARIMA is used as a baseline due to its established use in time-series analysis, but its linearity and stationarity assumptions limit its ability to capture complex outage dynamics. To address this, we include GRU and LSTM models, which learn nonlinear temporal dependencies from sequential data. We also evaluate the Transformer model, which uses attention mechanisms to capture long-range dependencies without recurrence.

For each county, data are split into training (80%) and validation (20%) sets. Performance is measured using Mean Squared Error (MSE), Mean Absolute Error (MAE), and R^2. The following sections describe model formulations, hyperparameter tuning, and cross-county performance comparisons.

ARIMA

Principle

ARIMA(p, d, q) models a series through autoregressive terms (p), differencing (d), and moving-average terms (q). Stationarity is checked via the Augmented Dickey-Fuller test: d = 0 when p-value < 0.05, otherwise d = 1. Models are fitted on log-transformed outages without exogenous features.

Hyperparameter Tuning

Hyperparameters are selected by grid search over p in {0,1,2}, d in {0,1}, q in {0,1,2}, choosing the combination with the lowest AIC. Walk-forward validation is then applied: after each one-step prediction, the true value is appended to the training history before the next forecast. Full-series fits also produce 24- and 48-hour forecasts.

Result Analysis

Results vary widely across counties. The best case is county 26063 (MSE 119.42, MAE 2.07, R-squared 0.60), where predictions track both minor fluctuations and larger spikes (Figure 6). The worst is county 26013 (MSE 4368.53, MAE 4.24, R-squared -0.19), where long zero-outage periods and occasional large spikes lead the model to underestimate extremes (Figure 7).

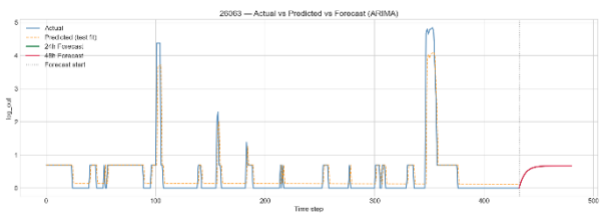

Figure 6 ARIMA prediction for county 26063

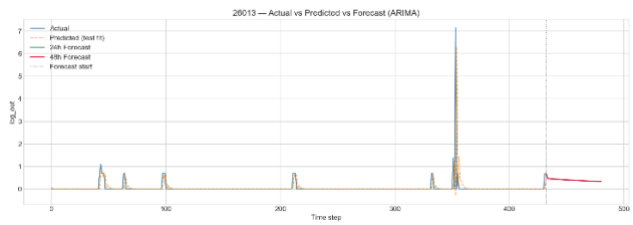

Figure 7 ARIMA prediction for county 26013

The overall averages are MSE 156,004.30, MAE 43.06, and R-squared 0.22. Several counties produce negative R-squared values, indicating that ARIMA is a reasonable baseline but struggles with irregular, weather-driven outage events.

GRU

Principle

The Gated Recurrent Unit (GRU) (Cho et al., 2014) is a recurrent neural network architecture designed to capture temporal dependencies in sequential data. Compared with LSTM, GRU uses a simplified gating structure with two gates:

1. Update gate, which controls how much of the previous hidden state is retained.
2. Reset gate, which determines how much past information contributes to the candidate's hidden state.

The candidate hidden state is generated from the current input and the reset-filtered previous state, and the final hidden state is obtained as a weighted combination of the previous state and the candidate state. Because GRU maintains a single hidden state and fewer parameters than LSTM, it often trains faster while achieving comparable performance.

Hyperparameter Tuning

The GRU network consists of two stacked GRU layers followed by a dense output layer. The loss function is MSE, and the maximum number of training epochs is 30.

The hyperparameters explored are: look_back(12, 24); gru_unit(32, 64); dropout(0.1, 0.2); batch_size(16, 32). Grid search is used to identify the best configuration for each county based on validation MSE.

Result Analysis

The GRU model generally performs better than ARIMA. The best performance is in county 26163, with 0.25 of MSE, 0.36 of MAE, 0.89 of R^2. The worst performance is in county 26003, with 0.1 of MSE, 0.17 of MAE, -0.15 of R^2. The average performance is 0.67 of MSE, 0.43 of MAE, 0.3 of R^2.

For County 26163, the model achieves excellent performance ($R^2 = 0.8935$). The predictions closely track both short-term fluctuations and a major outage event, demonstrating that GRU can learn complex temporal patterns. Figure 8 shows the prediction result for county 26163.

However, performance is weaker for counties with very sparse outages. For example, County 26003 has few events and low variance, making it difficult for the model to outperform a simple mean predictor, which leads to a negative R^2 value. Figure 9 shows the prediction result for county 26003.

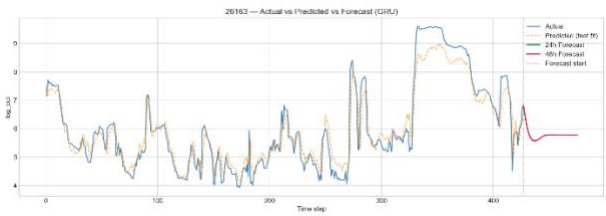

Figure 8 Prediction for County 26163 using GRU

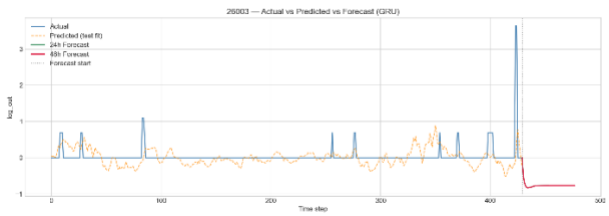

Figure 9 Prediction for County 26003 using GRU

Across all counties, GRU achieves an average R^2 of 0.2975, outperforming ARIMA and demonstrating its ability to capture nonlinear temporal patterns in outage data.

LSTM

Principle

The Long Short-Term Memory (LSTM) network (Hochreiter & Schmidhuber, 1997) is a recurrent architecture designed to address the vanishing gradient problem in traditional RNNs. Each LSTM unit contains four components:

1. Cell state, which stores long-term information.
2. Forget gate, which determines which information from the previous cell state should be discarded.
3. Input gate, which controls how new information is added to the cell state.
4. Output gate, which determines the hidden state output.

Through these gating mechanisms, LSTM can maintain long-term dependencies in sequential data and selectively update memory states during training.

Hyperparameter Tuning

The LSTM model uses two stacked layers and is trained for up to 30 epochs using MSE loss.

Hyperparameters explored include: look_back(12, 24); lstm_units(32, 64); dropout(0.1, 0.2); batch_size(16, 32). The optimal configuration is selected using grid search based on the lowest validation MSE.

Result Analysis

Performance varies across counties. For County 26019, the model achieves: MSE: 0.50, MAE: 0.38 and R^2: 0.59. The visualization of the prediction results is shown in the figure 10.

The prediction curve closely follows the overall trend of the outage series and captures most fluctuations during the validation period.

However, performance is weaker for some counties. For example, County 26073 achieves an R^2 of 0.0902, suggesting that the outage patterns are more irregular and difficult for the model to learn. The figure 11 shows the prediction result.

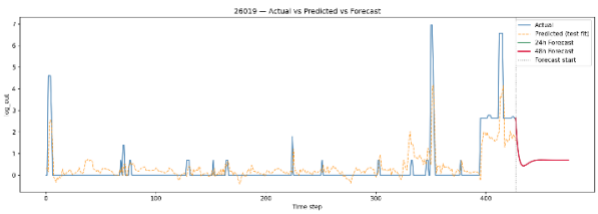

Figure 10 Prediction for County 26019 using LSTM

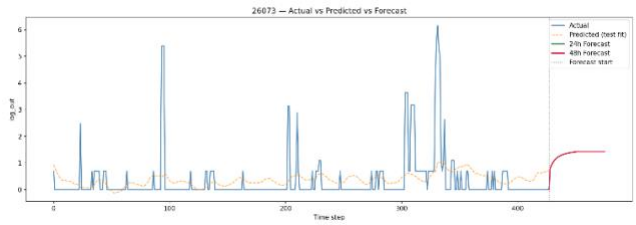

Figure 11 Prediction for County 26073 using LSTM

Overall, LSTM provides reliable predictions for many counties by leveraging its ability to capture long-term temporal dependencies.

Transformer

Principle

The Transformer (Vaswani et al., 2017) relies entirely on self-attention to model dependencies between all positions in a sequence simultaneously, without recurrence. Our architecture takes a window of W = 24 hours of features (historical log_out and weather variables) as input. Each time step is projected into a 64-dimensional embedding space, combined with learnable positional embeddings, and passed through 2 stacked encoder layers with multi-head self-attention and feed-forward networks. The full encoder output (24 × 64 = 1536 dimensions) is flattened and fed through a two-layer MLP prediction head that outputs 48 values—the predicted log_out for the next 48 hours.

Hyperparameter Tuning

A separate model is trained per county (83 total) with fixed hyperparameters: window_size = 24, d_model = 64, nhead = 4, num_layers = 2, dim_ff = 128, dropout = 0.15, batch_size = 16, lr = 5e-4, weight_decay = 1e-3, MSE loss. Input features include historical log_out, numeric weather variables, and temporal features (hour/day sine/cosine). We use the CosineAnnealingLR scheduler and early stopping (patience = 15). Features are standardized with StandardScaler; targets scaled with MinMaxScaler to [0, 1].

Result Analysis

We evaluate using MSE, MAE, and R^2 on single-step (t+1) prediction in log space on the validation set (last 20%). Top-performing county 26099 (MSE: 0.64, MAE:0.80, R^2: 0.28) with figure 12. The prediction curve generally follows the overall trend of the outage series, capturing several rises and drops in the signal, although the magnitude of some peaks is underestimated. This indicates that the Transformer is able to learn part of the temporal structure of the outage series, but still struggles to fully reproduce larger fluctuations.

Bottom-performing county 26007 because of near-constant outage values (MSE: 0, MAE: 0.01, R^2: 0) with figure 13.

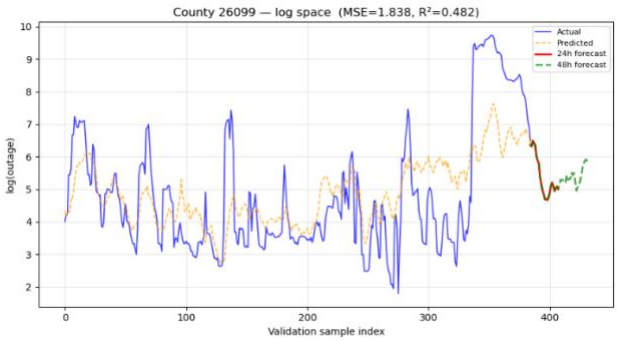

Figure 12 Prediction for County 26099

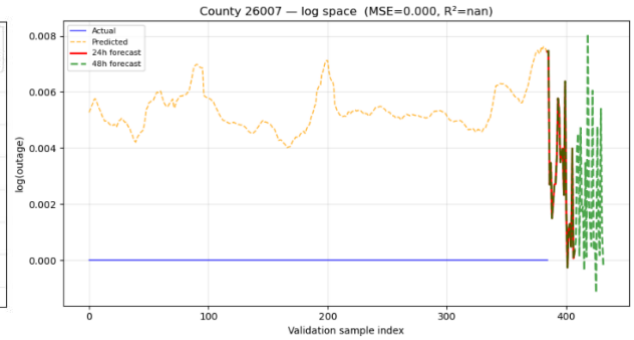

Figure 13 Prediction for County 26007

The average performance across all 83 counties is MSE: 1.61, MAE: 0.68 and R^2: -0.61.

Result

In this section, we compare the performance of the four models to determine the most suitable model for predicting power outages. To evaluate model performance, we compute the MSE, MAE, and R^2 across all counties. Table 1 presents the overall results.

Table 1 Model Performance Evaluation

Model	MSE	MAE	R^2
ARIMA	156004.30	43.06	0.2197
GRU	0.67	0.43	0.30
LSTM	0.70	0.45	0.34
Transformer	1.61	0.68	-0.61

To further compare the models, we rank each model based on the three evaluation metrics. For each metric, models are assigned a rank from 0 (best) to 3 (worst). The final score for each model is obtained by summing the ranks across the three metrics. The model with the lowest total rank is considered to have the best overall performance. Table 2 presents the ranking results and the corresponding sums. Based on this ranking method, LSTM achieves the lowest total score, indicating that it performs best among the four models.

Table 2 Model Ranking

Model	MSE	MAE	R^2	Sum
ARIMA	3	3	2	8
GRU	0	0	1	1
LSTM	1	1	0	2
Transformer	2	2	3	7

Among the models, the GRU model achieves the best performance, with an average MSE of 0.67, an average MAE of 0.43, and an average R^2 score of 0.30.

In addition to the numerical comparison, we also visualize the prediction results to provide further support for the comparison. As an example, we select County 26001. Figure 14(a-d) show the 24-hour and 48-hour forecasts, along with the actual and predicted values in the validation set for each model.

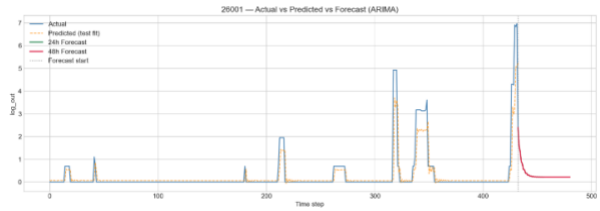

Figure 14(a) ARIMA Performance

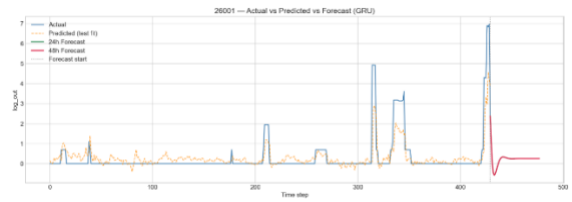

Figure 14(b) GRU Performance

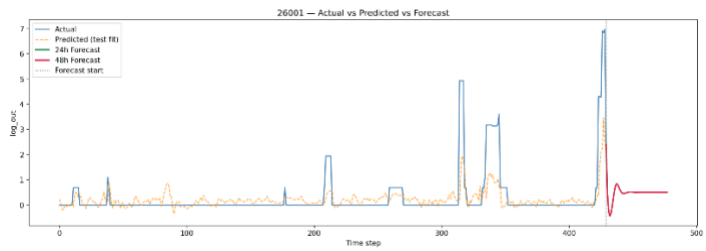

Figure 14(c) LSTM performance

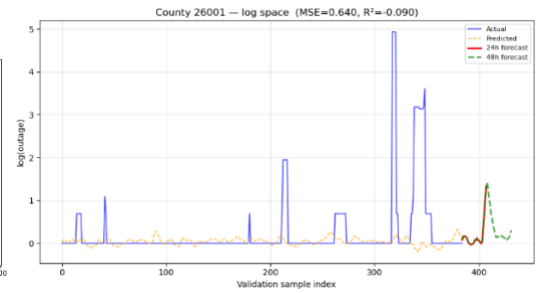

Figure 14(d) Transformer Performance

From these visualizations, we observe that the GRU model follows the trend of the actual outage values more closely than the other models, producing smoother and more accurate predictions. Overall, the results indicate that GRU is the most effective model for capturing the temporal patterns in the outage data, and therefore it is selected as the final model for prediction.

Power Stations Location Recommendation

Overview

This section optimizes the placement of 5 power stations across Michigan's 83 counties using historical outage data. Log outage values are inverse-transformed to recover actual demand, then averaged per county. Counties with demand above 1,000 are excluded. The 77 remaining counties total about 1263 demand units. Two algorithms are compared under a station capacity of 1000.

Algorithm 1: Capacitated Weighted K-Means

We run 30 random K-Means iterations (max 200 each). After each assignment step, overloaded clusters shed their most distant points to neighbors with spare capacity. Cluster centers are recomputed as demand-weighted centroids. The run with the lowest weighted inertia is kept. The station is built in the county nearest to the final centroid.

Algorithm 2: Greedy Capacitated Assignment

We seed the 5 stations at the 5 highest-demand counties, then assigns all remaining counties (sorted by demand, descending) to the nearest station with available capacity.

Results

Station	K-Means FIPS	K-Means Demand	K-Means Counties	Greedy FIPS	Greedy Demand	Greedy Counties
1	26081	255.62	33	26125	333.84	12
2	26125	698.1	13	26099	194.72	4
3	26021	68.05	6	26081	403.45	52

4	26065	224.95	14	26163	171.61	2
5	26013	16.24	11	26161	159.33	7

Comparison & Conclusion

K-Means produces more imbalanced loads. Greedy is faster and deterministic. Both methods exclude the same 2 over-capacity counties and leave no counties unassigned. Greedy is recommended for real deployment due to its larger coverage area. Figure 15 shows the station location with its assigned counties using a greedy algorithm.

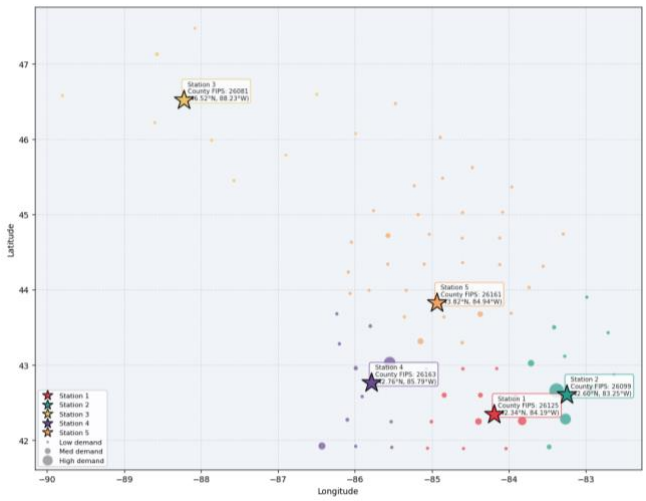

Figure 15 Power Stations Location

Policy Recommendation

Based on the GRU predictions and Greedy deployment results, we recommend the 5-station plan and discuss its policy implications below.

Our GRU model shows outage risk is highly concentrated in urban centers. The Greedy algorithm reflects this by seeding stations at the five highest-demand counties: Oakland (26125), Macomb (26099), Kent (26081), Wayne (26163), and Washtenaw (26161), four of which are in the Detroit metro area, with Kent (Grand Rapids) the lone western outlier. Station 3 (Kent) absorbs the largest load (403.45 across 52 counties) because it is the nearest station to most of western and northern Michigan, while Station 5 (Washtenaw) serves only 159.33 across 7 nearby counties. Demand-driven placement naturally prioritizes where generators are needed most.

The algorithm excludes 6 counties: 5 northern counties with effectively zero historical outages, and Saginaw (26145), whose demand of 1,101.86 exceeds the 1,000-household generator capacity. Saginaw is the key limitation of the 5-generator constraint and would require either multiple dedicated generators or permanent infrastructure upgrades such as tree trimming and powerline undergrounding. The MPSC has already moved in this direction (MPSC, 2025, 2026), and our county-level predictions could help prioritize which circuits to harden first.

The limitation is that our allocation is purely demand-based. It does not account for population vulnerability, critical facility locations, or restoration time differences. Demand estimates also rely on historical data, and shifting weather patterns under climate change may reshape future risk distributions.

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